

Rook to XSS: How I hacked chess.com with a rookie exploit

Rook to XSS: How I hacked chess.com with a rookie exploit - Jacob, Skii.dev.

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Content

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Playing Chess is one of the many hobbies I like to do in my spare time, apart from tinkering around with technology. However, I'm not very good at it, and after losing many games, I decided to see if I could do something I'm much better at; hacking the system!

This blog post is about how I used my cybersecurity knowledge to find XSS on the #1 chess site on the internet, with a user base of over **100 million** members - [Chess.com](https://chess.com). But first, a bit of preface (that includes a slightly less serious, although amusing, OSRF vulnerability!)

In early 2023, I started playing a lot on chess.com; during a discussion with a friend on Discord, I persuaded them to also sign up to the site and used a feature offered by chess.com to become friends once they signed up immediately.

!

This feature reminded me of the [MySpace worm](#) in ~2005 (heck, I wasn't even alive then!) when Samy Kamkar injected some code into his profile that would friend anyone who visited it and then inject the same code into their profile (hence creating the worm). I wondered if it would be possible to do something similar here. I clicked on the link and created a new account, then checked the dev tools network tab - interestingly, **after** the account had been made, it sent a GET request to <https://chess.com/registration-invite?hash=XXX>

!

This meant that if I could get a user to request this URL, it would force them to friend me automatically. Coincidentally, I was also messing with my settings when I came across the holy grail.....a TinyMCE rich text editor with an image upload function!

!

Let's see what happens when I insert a [link](#) for the image. Is the URL embedded directly, or will there be some safe handling to protect against request forgeries?

!

Chess.com handles this server-side by re-uploading the image to their content hosting server and then pointing the image URL to that. Hmm...*But* what about using a link whose root domain is `chess.com`? Would that still get re-uploaded? This would be important, especially when chained with a specific URL like...

!

!

BINGO! I switched to my alt account, navigated to my main account's profile and then checked my alt's friend list - it had successfully added my main account.

!

I genuinely couldn't believe this had worked, and I had quite a chuckle about it. During the bug-bounty report & triage, the developers tried to implement a block because when I tried to reproduce it again for them, it came up with the following error message:

!

No problem at all, though; I managed to bypass it again by setting up a subdomain that included `chess.com` & redirected it to `/registration-invite`

!

And here's what it looked like when visiting my profile:

!

After finding and reporting this bug, I was very interested in what else I could achieve by abusing TinyMCE - could I get XSS? How good was the sanitisation? That brings us to the exciting part of the blog post...

After playing around with the editor for a bit, I realised I wouldn't get very far without using something like [Burp's proxy](#) to intercept the request to save my `About` description and inject raw HTML code directly (Anything written in the editor was treated as text). Of course, as one would expect, there were already preventions to remove any non-whitelisted attributes and tags - so let's look at what *is* allowed.

Viewing the TinyMCE config on the site (you can find this in the `tinymce-lazy-client.js` file), for `img` tags the `background-image` style attribute is in the `Allow` list and does not get filtered out. This was a long shot, but I wondered if the function to re-upload any external URL for the image also got applied to the attributes. Well...no harm in trying; let's see what happens using the following payload:

```
<p>payload here</p>',
    'profile[save]': '',
    'profile[_token]': 'xxxx',
}

while True:
    data['profile[about]'] = input()

    response = requests.post('https://www.chess.com/settings', cookies=cookies, headers=headers, data=data)
    print(response)
```

Disclaimer: This process was done under CC's bug bounty program, strictly staying within the scope; PII information has been censored by request. This bug was reported over a year ago and has passed the 9-month disclosure time. Please only hack on sites that you have written permission to!